Department of Electrical and Electronics Engineering
Academic Year 2023 – 2024 (Even Semester)
Degree, Semester & Branch: VI Semester B.E. EEE
Course Code & Title: EE3035 Grid Integrating Techniques and Challenges
Name of the Faculty member (s): Mrs. S. Jeyanthi

Innovative Practice Description

- **Unit / Topic: Unit I / Power Plants India**
- **Course Outcome:** CO 1
- **Topic Learning Outcome:** 1e
- **Activity Chosen:** One minute paper
- **Justification:**

  India's power plants include a diverse mix of energy sources such as coal, natural gas, hydroelectric, nuclear, and renewable sources like solar and wind. This energy infrastructure is crucial for supporting the country's rapidly growing economy and expanding population. A one-minute paper is used for this concept because it prompts students to quickly summarize key points and reflect on their understanding of India's power plants. This technique helps in identifying any areas of confusion and reinforces learning by encouraging concise and focused thinking.

  Time Allotted for the Activity: 5 minutes
- **Details of the Implementation:**
  At the end of the class, students were asked to write about the topic discussed in the class. The students expressed the understood content and the content which were not clear in that particular topic. This activity shows whether the students can able to understand the specific topic and their involvement the particular class.
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO 1	3	-	-	-	-	-	-	-	-	1	-	1

(1 – Low 2 – Moderate 3 – High)

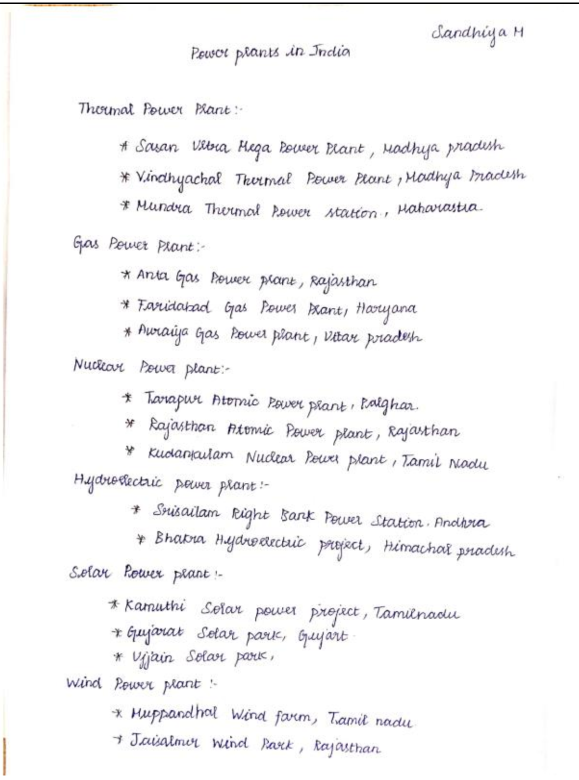
- **PO / PSO mapped:**

	PO9
Innovative practice	1
Justification for correlation	The students can Function effectively as an individual

- **Images / Screenshot of the practice:**

Innovative Teaching Method Execution

Power Plants India – One minute paper



Sandhiya M

Power plants in India

Thermal Power Plant :-

* Sasan Ultra Mega Power Plant, Madhya pradesh
* Vindhyachal Thermal Power Plant, Madhya Pradesh
* Mundra Thermal Power station, Maharastra.

Gas Power Plant :-

* Anta Gas Power plant, Rajasthan
* Faridabad Gas Power Plant, Haryana
* Auraiya Gas Power plant, Uttar pradesh

Nuclear Power plant :-

* Tarapur Atomic Power plant, Palghar.
* Rajasthan Atomic Power plant, Rajasthan
* Kudankulam Nuclear Power plant, Tamil Nadu

Hydroelectric power plant :-

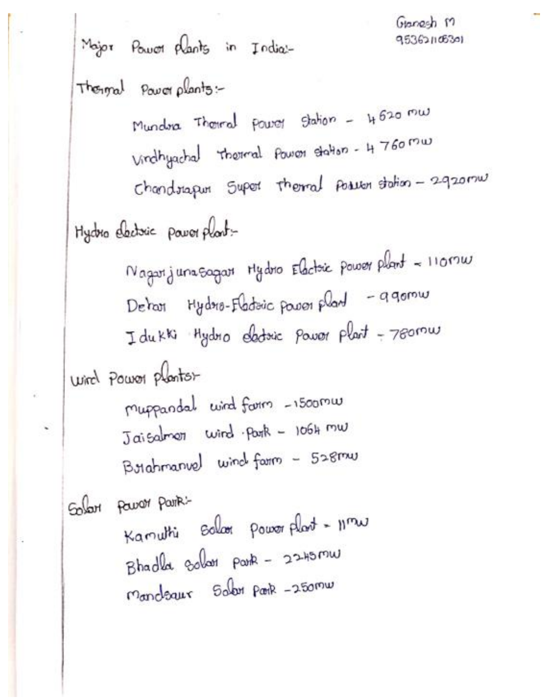
* Srisailam Right Bank Power Station. Andhra
* Bhakra Hydroelectric project, Himachal pradesh

Solar Power plant :-

* Kamuthi Solar power project, Tamilnadu
* Gujarat Solar park, Gujart
* Ujjain Solar park,

Wind Power plant :-

* Muppandhal Wind farm, Tamil nadu
* Jaisalmer Wind Park, Rajasthan



Ganesh M
953621106301

Major Power Plants in India:-

Thermal Power plants:-

Mundra Thermal power station - 4620 mw
Vindhyachal thermal Power station - 4760 mw
Chandrapur Super thermal Power station - 2920mw

Hydro electric power plant:-

Nagarjunasagar Hydro Electric power plant - 110mw
Dehar Hydro-Electric power plant - 990mw
Idukki Hydro electric Power plant - 780mw

wind Power plants:-

muppandal wind farm - 1500mw
Jaisalmer wind Park - 1064 mw
Brahmanvel wind farm - 528mw

Solar power Park:-

Kamuthi Solar Power plant - 11mw
Bhadla Solar park - 2245mw
mandsaur Solar Park - 250mw

- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***
    - ✓ Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

  - ***Benefit of the practice:***
    - ✓ Students can able to attend the question even in the questions are in indirect form.
    - ✓ Students can able to explain the concepts in examination without any confusion.
  - ***Challenges faced in implementation:***
    - ✓ Time utilization for conducting activity.

References:

1. Integration of Renewable Energy Sources with Smart Grid, M. Kathiresh, A. Mahaboob Subahani, and G.R. Kanaga chidambaresan, Scrivener & Wiley, 2021, 1st Edition.
2. Control and Operation of Grid-Connected Wind Energy Systems, Ali M. Eltamaly, Almoataz Y. Abdelaziz, Ahmed G. Abo-Khalil, Springer 2021, 1st Edition.

Signature of Faculty Member **HOD**

Department of Electrical and Electronics Engineering
Academic Year 2023 – 2024 (Even Semester)
Degree, Semester & Branch: VI Semester B.E. EEE
Course Code & Title: EE3035 Grid Integrating Techniques and Challenges
Name of the Faculty member (s): Mrs. S. Jeyanthi

Innovative Practice Description

- **Unit / Topic: Unit II / Transformers in Electric Grid**
- **Course Outcome:** CO 2
- **Topic Learning Outcome:** 2d
- **Activity Chosen:** Think pair share
- **Justification:**
  - ✓ It helps students to think individually about a topic or answer to a question.
  - ✓ It teaches students to share ideas with classmates and builds oral communication skills.
  - ✓ It helps focus attention and engage students in comprehending the reading material.
- **Time Allotted for the Activity:** 10 minutes
- **Details of the Implementation:**

  Think-Pair-Share innovative practice conducted for VI year ECE students, after explained the concept of Transformers in Electric Grid. First, I asked each student to think about the role of transformers in the electric grid for 2 minutes. They should consider their types, working principles, and significance in power distribution. Then I make them as a pair up to discuss their thoughts and answers. Encourage them to compare their ideas and clarify any misunderstandings. 3 minutes. Finally I invited the pairs to share their insights with the larger group or the whole class for 10 minutes.
-
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO 2	3	3	2	2	3	-	-	-	-	1	-	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO9**
	1
Justification for correlation	The students can Function effectively as a team

- **Images / Screenshot of the practice:**

Innovative Teaching Method Execution
Transformers in Electric Grid – Think Pair Share

K. Govindharaj
M. Udhiya Kumar.

Transformers in Power Grid:-

The Transformers are used to Transfor the Electricity. It Plays the major role in Power grids. by enabling efficient transmission and distribution of Power.

The major Role of Transformer in Power Grids are:

i) Voltage Transmission.
ii) Grid Stability and control.
iii) Isolation and Safety.
iv) Phase Shifting.
v) Fault management and protection.

The main Role is Step Up and Step down the Voltage for Transmission and distribution. The Transformers are help to maintain Voltage Stability and Power quality with in the Grid. Transformer incorporate devices Such as Circuit Breaker and fuses to detect and isolate faults quickly. This helps in minimizing distruptions in Power Supply and protect the Equipments from damage during the faults conditions.

- **Reflective Critique:**
  - ❖ ***Feedback of practice from students and other stakeholders:***
    - ✓ Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

❖ ***Benefit of the practice:***

Think-pair-sharing forces all students to attempt an initial response to the question, which they can then clarify and expand as they collaborate. It also gives them a chance to validate their ideas in a small group before mentioning them to the large group, which may help shy students feel more confident participating.

❖ ***Challenges faced in implementation:***

I planned the activity for 10 minutes. But in Class room it takes 15 minutes.

References:

1. Integration of Renewable Energy Sources with Smart Grid, M. Kathiresh, A. Mahaboob Subahani, and G.R. Kanaga chidambaresan, Scrivener & Wiley, 2021, 1st Edition.
2. Control and Operation of Grid-Connected Wind Energy Systems, Ali M. Eltamaly, Almoataz Y. Abdelaziz, Ahmed G. Abo-Khalil, Springer 2021, 1st Edition.

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Course Code & Title: EE3035 Grid Integrating Techniques and Challenges
Name of the Faculty member (s): Mrs. S. Jeyanthi

Innovative Practice Description

- **Unit / Topic: Unit III / The Sizing of DC/DC Converters for Micro grid**
- **Course Outcome:** CO 3
- **Topic Learning Outcome:** 3e
- **Activity Chosen:** Strip sequence
- **Justification:**

The sizing of DC/DC converters for a microgrid with strip sequence involves determining the appropriate specifications and configurations of these converters to ensure efficient power conversion and distribution within the microgrid. This activity helps students apply what they have learned through reading or didactic teaching. This approach can strengthen students' logical thinking processes and test their mental model of a process. The activity can be done in pairs or groups.

- **Time Allotted for the Activity:** 10 minutes
- **Details of the Implementation:**

Strip Sequence activities require students to actively engage in their learning, often by connecting their prior knowledge to new information. When creating a Strip Sequence, a student frequently interacts with a textbook, notes from class, an instructor, classmate, or study group. In my subject Mini map was conducted for the Sizing of DC/DC converters for Micro grid. The students are actively participated in this event.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO 5	3	3	3	2	3	-	-	-	-	1	-	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO9
	1
Justification for correlation	The students can Function effectively as a team

- **Images / Screenshot of the practice:**

Innovative Teaching Method Execution

The Sizing of DC/DC Converters for Micro grid – Types

methodology / Strip Sequence

G. Angalaeswari

Sizing of DC-DC converter in a microgrid

Step1:- To determine load requirement:- By calculating the total power demand of all DC loads in the microgrid. This involves identifying the maximum power each load will draw.

Step 2:- To consider voltage level :- By calculating the total power demand of all DC loads in the microgrid. for the input and output voltage levels.

Step3:- To account for efficiency:- DC-DC converters have efficiency rating with vary load and input/output voltage level. Efficiency range 80% and 95%.

Step 4:- To calculate power rating:-

Converter power rating = $\frac{\text{Total power demand}}{\text{Converter Efficiency}}$

Step5:- To dynamic loads:- Microgrids often have dynamic loads, where the power demand can fluctuate.

Step 6:- To Temperature considerations:- Evaluate the ambient temperature conditions where DC-DC converter

Step7:- To select the type of DC-DC converter:- Choose the type best matches the voltage and efficiency.

Step8:- To manufacturer and model selection:- Research different manufacture and model.

Step 9:- To Design Margin:- A typical margin might be around 20% above calculate power

Step 10:- To implement and testing:- Test the converter under various load.

- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***
  - Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

  - ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

The benefits of Strip Sequence are a great way for students to make notes on all of the information they receive. It made the students the general steps involved in the process and make the students to implement the procedure for the particular topic.

- ***Challenges faced in implementation:***

Normally teachers will give longer explanations in the notes section of the topic. The students are made into groups and to arrange the strip of process for the topic of Energy Audit. I planned the activity for 10 minutes only. But in real scenario it takes 15 minutes.

References:

1. Integration of Renewable Energy Sources with Smart Grid, M. Kathiresh, A. Mahaboob Subahani, and G.R. Kanaga chidambaresan, Scrivener & Wiley, 2021, 1st Edition.
2. Control and Operation of Grid-Connected Wind Energy Systems, Ali M. Eltamaly, Almoataz Y. Abdelaziz, Ahmed G. Abo-Khalil, Springer 2021, 1st Edition.

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Innovative Practice Description

- **Unit / Topic: Unit IV / Importance of Custom Power Devices- Power Quality Point of View**
- **Course Outcome:** CO 4
- **Topic Learning Outcome:** 4d
- **Activity Chosen:** Mini map
- **Justification:**

  Mini mapping is a creative way to set goals, solve problems and design action plans. It quickly records ideas in a free-form way. When groups use mini mapping, the thoughts of each participant easily trigger ideas in others. This dynamic group interaction encourages breaking free of old patterns to uncover new and innovative approaches. For this topic, there are different types of magnetic material and its characteristics can be brought in to single map for better understanding.
- **Time Allotted for the Activity:** 10 minutes
- **Details of the Implementation:**

  Mini mapping activities require students to actively engage in their learning, often by connecting their prior knowledge to new information. When creating a mini map, a student frequently interacts with a textbook, notes from class, an instructor, classmate, or study group.
  In my subject Mini map was conducted for the topic Importance of Custom Power Devices- Power Quality Point of View. The students are voluntarily drawn the map.
- **CO – PO / PSO mapping:**

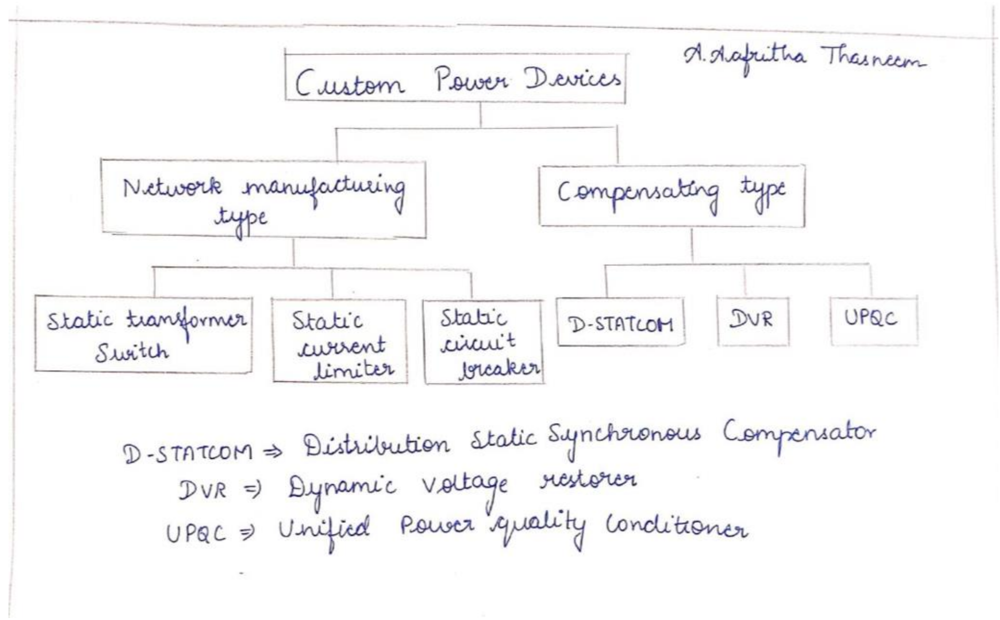
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO4	3	3	1	2	3	-	-	-	-	1	-	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO9**
	1
Justification for correlation	The students can Function effectively as a team

- **Images / Screenshot of the practice:**

Innovative Teaching Method Execution
Importance of Custom Power Devices- Power Quality Point of View **-** Mini map


- **Reflective Critique:**
  - ❖ ***Feedback of practice from students and other stakeholders:***

    Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

  - ❖ ***Benefit of the practice:***

    The benefits of Mini Maps are a great way for students to make notes on all of the information they receive. It helps the students to note down only the most important information using key words, and then make connections between facts and ideas visually – keeping all of your topic thoughts together on one sheet. It made key note making easier to students, as it reduces pages of notes into one single side of paper. Also mini map made slow learners to remember the information more quickly.

Challenges faced in implementation:

Normally teachers will give longer explanations in the notes section of the topic. The students are made into groups and to draw the map to Indicate relationships between the topics in biomass conversion process. I planned the activity for 10 minutes only. But in real scenario it takes 15 minutes to complete this activity.

References:

1. Integration of Renewable Energy Sources with Smart Grid, M. Kathiresh, A. Mahaboob Subahani, and G.R. Kanaga chidambaresan, Scrivener & Wiley, 2021, 1st Edition.
2. Control and Operation of Grid-Connected Wind Energy Systems, Ali M. Eltamaly, Almoataz Y. Abdelaziz, Ahmed G. Abo-Khalil, Springer 2021, 1st Edition.

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Innovative Practice Description

- **Unit / Topic:** Unit V / Grid integration of WECS - Challenges in the Integration of WECS
- **Course Outcome:** CO5
- **Topic Learning Outcome:** 5b
- **Activity Chosen:** Flipped Classroom
- **Justification:**
  This topic is chosen, since it can be easily understood by students. Students can improve their understanding on this topic by this activity. More time for discussion is given the class unlike traditional learning. This activity greatly helps the student to know the concept much deeper.
- **Time Allotted for the Activity:** 25 Minutes
- **Details of the Implementation:**
  Each team was asked to do literature survey about the given topic. They should include at least 5 Reference material about the given topic for which the material is already shared. Students in a team were asked to present their topics. Queries will be asked by the students to the presenting team
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO 1	3	3	2	2	3	-	-	-	-	1	-	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO9**
	1
Justification for correlation	The students can Function effectively as an individual

- **Images / Screenshot of the practice:**

Innovative Teaching Method Execution
Grid integration of WECS - Challenges in the Integration of WECS– Flipped Classroom


- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:*** The students enjoyed this Flipped classroom Activity. After the class many students requested me to insist other Course faculty members to conduct this activity. I have taken this feedback to by Head of the Department
  - ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)
    The students actively participated in the flipped classroom activity. As a teacher I could know the real students feeling, their understanding level from this activity. This greatly helped me to clear all the student doubt and make them more involved in the learning process
  - ***Challenges faced in implementation:*** Still few students were hesitant to convey their points and queries. Few students have not taken this activity seriously and remain quite in the class

References:

1. Design of smart power grid renewable energy systems, Third Edition, Ali Keyhani, Wiley 2019.
2. https://www.wind-energy-the-facts.org/images/chapter2.pdf

Signature of Faculty Member **HOD**